Comprehensive Multiphysics Simulation and Performance Optimization of MoS_2 Field-Effect Transistors Using High-k and Ferroelectric Dielectrics for Scaled Logic and Memory Applications

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Abstract

The continuous miniaturization of semiconductor devices has catalyzed a transformative evolution in the field of nanoelectronics. However, conventional silicon-based field-effect transistors (FETs) are increasingly hindered by fundamental physical limitations as scaling approaches sub-5 nm nodes. This impending saturation in CMOS technology necessitates the pursuit of alternative materials and device architectures that can offer both superior electrical performance and physical scalability. Among the myriad options being explored, two-dimensional (2D) transition metal dichalcogenides (TMDs) have emerged as leading candidates, owing to their atomically thin structures, excellent gate control, and intrinsic bandgaps. In particular, molybdenum disulfide (MoS_2) offers a promising path forward due to its favorable band structure, compatibility with existing fabrication methods, and stability under ambient conditions. This paper presents an extensive multiphysics simulation study of MoS_2-based FETs using finite element modeling tools to understand, predict, and optimize their performance characteristics. Both monolayer and multilayer channel configurations are explored under various gate dielectric conditions, including traditional SiO_2, high-k Al_2O_3, and ferroelectric HfO_2. The simulations provide a self-consistent solution to the Poisson and drift-diffusion equations, enabling precise insight into charge transport, electric potential profiles, recombination dynamics, and memory effects. Particular focus is given to the electrostatic benefits of high-k dielectrics, the subthreshold behavior under aggressive scaling, and the hysteresis characteristics introduced by ferroelectric polarization. The results demonstrate the feasibility of MoS_2 transistors for logic and memory applications, achieving sub-60 mV/dec subthreshold swing, memory windows near 0.8 V, and high on/off current ratios. Furthermore, the simulations align closely with experimental data from recent literature, reinforcing the accuracy and predictive capability of the finite element framework. The outcomes offer clear guidelines for optimizing channel thickness, dielectric stack composition, and gate geometry to achieve robust, low-power 2D FETs, marking a significant step toward future nanoscale electronics.

Index Terms

MoS_2 FET, 2D materials, negative capacitance, COMSOL simulation, ferroelectric HfO_2, high-k dielectric, subthreshold swing, threshold voltage, nanoscale electronics, memory FET.

I. INTRODUCTION

The relentless advancement of semiconductor technology, long driven by Moore's Law, is now encountering fundamental limits as device scaling approaches atomic dimensions. For decades, reducing transistor dimensions has led to enhanced performance, lower power consumption, and increased integration density. However, the scaling of silicon-based metal-oxide-semiconductor field-effect transistors (MOSFETs) beyond the 7 nm node has revealed serious physical constraints that cannot be mitigated through traditional engineering solutions alone [1], [2]. These include short-channel effects (SCEs), such as drain-induced barrier lowering (DIBL), increased gate leakage due to ultrathin dielectrics, subthreshold swing (SS) limitations due to thermionic emission, and reliability concerns under high-field operation [3].

Silicon, despite being the foundation of modern electronics, is beginning to reach its performance ceiling in ultra-scaled applications. As the gate length is pushed toward sub-5 nm dimensions, control over the channel becomes increasingly difficult. Even advanced architectures such as FinFETs and gate-all-around FETs (GAAFETs) suffer from quantum confinement issues, parasitic capacitance, and power density challenges. The standard 60 mV/decade SS limit, dictated by thermal emission in conventional FETs, prevents further lowering of the operating voltage without sacrificing the ON-current or switching speed [4]. Furthermore, continued reduction of the oxide thickness leads to unacceptable levels of tunneling leakage, making it imperative to explore both new materials and new mechanisms of operation.

In response to these challenges, two-dimensional (2D) materials have emerged as promising alternatives for future transistor channels. Among the diverse family of 2D materials, molybdenum disulfide (MoS_2), a member of the transition metal dichalcogenides (TMDs), offers a combination of favorable properties for nanoscale transistors. MoS_2 possesses an intrinsic and tunable bandgap that transitions from indirect (bulk) to direct (monolayer), which is crucial for digital logic and low-power applications [5]. Its monolayer form exhibits a direct bandgap of approximately 1.8 eV, enabling high ON/OFF ratios essential for switching performance. Additionally, MoS_2 provides high mechanical flexibility, strong electrostatic gate control due to its atomic thickness, and excellent compatibility with standard semiconductor processes.

Compared to graphene, which lacks a bandgap and thus suffers from poor current saturation, MoS_2 enables full channel depletion and distinct OFF states, making it far more suitable for FET-based logic circuits [6]. Moreover, MoS_2 is compatible with various synthesis techniques, including mechanical exfoliation, chemical vapor deposition (CVD), and atomic layer deposition (ALD), allowing for scalable fabrication and integration with conventional back-end-of-line (BEOL) processing [7]. This compatibility opens the door to large-area, cost-effective production of MoS_2-based FETs for both standalone logic and embedded memory functions.

One of the most promising directions in MoS_2 FET development is the integration of high-permittivity (high-k) and ferroelectric dielectrics into the gate stack. Materials such as aluminum oxide (Al_2O_3) and hafnium oxide (HfO_2) offer superior gate control by enhancing gate capacitance without requiring ultrathin layers, thereby reducing gate leakage. Ferroelectric variants of HfO_2, made possible through appropriate doping and thermal processing, provide negative capacitance effects, which can effectively amplify the internal gate voltage. This leads to steeper SS values below the thermionic limit, offering significant reductions in switching energy [8]. In addition, ferroelectric layers introduce bistable polarization states that can be exploited for memory operation, enabling dual functionality in a single device platform.

Despite these advantages, optimizing MoS_2-based FETs requires precise understanding of the interplay between channel material properties, gate dielectric selection, and device geometry. Experimental investigation of all these variables is time-consuming and cost-prohibitive. As a result, simulation-driven approaches have become invaluable tools for exploring the performance limits and design trade-offs of next-generation transistors. Finite element modeling (FEM), particularly through platforms such as COMSOL Multiphysics, provides a comprehensive and customizable environment for simulating electrostatics, charge transport, carrier recombination, and ferroelectric behavior within a single unified framework [9].

In this study, we employ a detailed FEM approach to simulate MoS_2 FETs across a range of configurations, including monolayer and multilayer channels, top-gated and back-gated architectures, and gate dielectrics comprising SiO_2, Al_2O_3, and ferroelectric HfO_2. The models are implemented in a two-dimensional (2D) domain with appropriate meshing, boundary conditions, and material parameters derived from experimental data. The simulation platform solves the Poisson equation coupled with the drift-diffusion equations for carrier transport, along with the Landau–Khalatnikov equation for polarization dynamics in ferroelectric simulations.

A particular focus is placed on examining the electrical characteristics of the MoS_2 FETs under varying gate biases. Key performance metrics such as threshold voltage (Vth), subthreshold swing (SS), ON/OFF current ratio, recombination rate, and energy band profiles are extracted and analyzed. In ferroelectric configurations, hysteresis loops are modeled and evaluated for their potential in memory and steep-slope logic applications. A total of 23 figures from the simulation output are presented and explained in detail, providing a comprehensive view of both the operational behavior and physical phenomena inside the device structures.

This paper is structured as follows: Section II presents a background review of recent advancements in MoS_2 FETs, highlighting challenges in scaling, material integration, and prior simulation efforts. Section III outlines the simulation setup, including device geometry, material parameters, boundary conditions, and meshing strategy. The detailed results and physical interpretations are presented in Section IV, and optimization insights are discussed in Section V. Section VI concludes the work with a summary of key findings and directions for future research.

II. BACKGROUND AND RELATED WORK

Over the past several decades, the progress of semiconductor technology has been guided by continuous device scaling, which has enabled higher integration density, improved performance, and reduced power consumption. However, as traditional silicon-based MOSFETs approach sub-5 nm technology nodes, fundamental physical limitations have become unavoidable. Challenges such as increased leakage current, reduced gate control, and short-channel effects (SCEs) have emerged as critical bottlenecks in the further scaling of CMOS devices [6], [7].

Among the most notable limitations is the subthreshold swing (SS), which determines how effectively a transistor can switch from the OFF state to the ON state. For conventional thermionic FETs, the minimum achievable SS at room temperature is 60 mV/decade due to the Boltzmann transport limit [8]. This restricts the ability to lower the supply voltage, as doing so results in an exponential rise in the OFF-state current. Attempts to overcome this by reducing gate oxide thickness have led to excessive tunneling currents and reliability concerns. New materials and novel device concepts are thus required to sustain low-power, high-performance computing in the future.

To this end, two-dimensional (2D) materials — especially transition metal dichalcogenides (TMDs) such as MoS_2 — have been proposed as alternatives to silicon in next-generation transistors. Their atomically thin geometry allows for near-ideal gate control, even at channel lengths below 10 nm [9]. Unlike graphene, which has no bandgap, MoS_2 features a direct bandgap of ~1.8 eV in monolayer form and an indirect gap (~1.2 eV) in multilayer configurations, making it suitable for logic applications [10]. Additionally, the absence of dangling bonds on its surface reduces interface trap densities, improving electrostatics and enabling cleaner contact formation. Another significant advantage of MoS_2 is its compatibility with advanced dielectric materials. Traditional gate dielectrics such as SiO_2 offer good insulation but suffer from low dielectric constants ($\kappa \approx 3.9$), limiting their gate capacitance unless scaled to ultra-thin dimensions. High-k dielectrics like Al_2O_3 ($\kappa \approx 9$) and HfO_2 ($\kappa \approx 20$–25) provide better electrostatic control while reducing gate leakage [11]. When these dielectrics are integrated with MoS_2, they improve the overall device performance by increasing capacitance, reducing interface states, and enabling more aggressive scaling of equivalent oxide thickness (EOT).

More recently, the ferroelectric phase of HfO_2 has garnered attention for its ability to induce negative capacitance behavior when used as a gate dielectric in FETs. This effect occurs due to the internal energy profile of ferroelectric materials, which can amplify the gate voltage through polarization switching. The result is a sub-thermal subthreshold swing (<60 mV/decade), which directly translates to reduced power consumption and faster switching times [12]. Furthermore, the bistable nature of ferroelectric polarization can be harnessed to create non-volatile memory within the same device, opening up the possibility for unified logic and memory architectures.

Experimental studies on MoS_2 transistors with ferroelectric HfO_2 have demonstrated promising characteristics. Devices have achieved steep SS, high ON/OFF current ratios, and hysteresis loops indicative of ferroelectric switching. These attributes confirm the feasibility of using MoS_2 as a channel material in both logic and memory applications, particularly when combined with advanced dielectric engineering [13]. However, understanding the internal mechanisms that govern these effects—such as polarization saturation, recombination dynamics, and electrostatic field distribution—requires more than experimental observation alone.

This is where simulation and modeling become crucial. Finite element simulation platforms, especially COMSOL Multiphysics, have emerged as powerful tools for analyzing semiconductor devices. By solving the coupled Poisson, drift-diffusion, and carrier continuity equations, these platforms provide insights into electric potential profiles, carrier transport, and recombination-generation behavior inside nanoscale transistors [14]. The ability to incorporate ferroelectric material models, such as the Landau–Khalatnikov (L-K) equation for polarization dynamics, further enhances the simulation's capability to reflect real-world physics.

While several modeling efforts have examined specific aspects of MoS_2 FETs—such as channel thickness effects, dielectric constant variation, or hysteresis characteristics—few have combined these into a single, comprehensive framework. In most cases, simulations focus only on current-voltage (I–V) curves or threshold voltage tuning, ignoring critical internal mechanisms such as band edge modulation, recombination rate mapping, and memory retention behaviors under varying gate and drain voltages. Moreover, few studies simulate both monolayer and multilayer MoS_2 configurations across different dielectric environments (SiO_2, Al_2O_3, HfO_2), particularly with ferroelectric switching.

In addition, it is important to recognize that multilayer MoS_2 devices exhibit distinct characteristics from their monolayer counterparts. For instance, multilayers typically show higher mobility due to reduced surface scattering but also suffer from a reduced bandgap and increased interlayer coupling. These trade-offs influence key performance metrics such as drive current, SS, and threshold voltage, and must be accounted for in any predictive modeling strategy [15].

This study addresses these gaps by presenting a comprehensive simulation framework that integrates multilayer MoS_2 channels with different dielectric materials, including ferroelectric HfO_2, under varied gate and drain biases. The model captures essential behaviors such as subthreshold swing, hysteresis, recombination activity, and energy band dynamics, offering a holistic view of device performance. Through extensive simulation outputs and full-paragraph figure explanations, we aim to provide both quantitative results and qualitative understanding to guide the future design and optimization of MoS_2-based transistors.

III. SIMULATION METHODOLOGY

A. Overview of Simulation Framework

All simulations in this study were performed using COMSOL Multiphysics, an advanced finite element analysis platform capable of solving complex, multiphysics equations with high numerical precision. The simulations focused on analyzing the electrical behavior of MoS_2 field-effect transistors (FETs) under a variety of physical configurations, including variations in channel thickness (monolayer to multilayer), dielectric stack material (SiO_2, Al_2O_3, and ferroelectric HfO_2), and bias conditions (top-gate and back-gate voltage sweeps). Each simulation was executed using the Semiconductor Module and included the drift-diffusion equations for electron and hole transport, the Poisson equation for electrostatic potential, and the Landau–Khalatnikov model for ferroelectric polarization behavior when applicable.

Device Structure and Geometry **Fig. 1 (a)** illustrates the baseline monolayer MoS_2 FET 2D schematic architecture modeled in this work. The device consists of a single-layer MoS_2 channel (thickness 0.65 nm) placed atop a degenerately doped silicon substrate acting as a bottom gate, separated by a 30 nm SiO_2 gate dielectric. Source and drain contacts are modeled as ideal ohmic Scandium/Au electrodes (height: 3 nm, width: 8 nm) defined at the channel edges. Dual-gate configurations introduce a top gate electrode above the MoS_2 channel, isolated by a high-κ dielectric layer—Al_2O_3 (14 nm thickness) or ferroelectric HfO_2 (30 nm thickness). Multilayer devices replicate this geometry with four stacked MoS_2 layers. Channel lengths (L) of 50 nm and 100 nm and channel widths (W) of 4 µm are considered to investigate short-channel scaling effects.

Physics Models and Boundary Conditions The semiconductor physics interface in COMSOL solves the coupled Poisson and drift-diffusion equations:

Poisson's equation: $\nabla\cdot(\varepsilon\varepsilon_0\nabla V) = -q(p - n + ND - NA)$ – Electron/hole continuity: $\partial n/\partial t = (1/q)\nabla\cdot Jn + G - R$, $Jn = q\mu n\, n\, \nabla V + qDn\, \nabla n$ – Similar expressions for holes. Carrier recombination is modeled using Shockley–Read–Hall (SRH) with lifetimes $\tau n = \tau p = 1.5$ ns [51]. Contacts are defined as Dirichlet boundaries with fixed potential (source at 0 V, drain at VDS) and Ohmic injection; gates use floating potential conditions governed by external bias (VG). This graphical **abstract 1 (b)** presents a formal overview of the multiphysics simulation and optimization of MoS_2-based field-effect transistors (FETs) for advanced logic and memory applications. The schematic depicts the simulated device architecture, incorporating a multilayer substrate, monolayer MoS_2 channel, and top-gate dielectrics including Al_2O_3 and ferroelectric HfO_2. Using COMSOL Multiphysics, the modeling approach integrates electrostatics (via Poisson's equation), carrier transport (drift-diffusion), and polarization dynamics (Landau–Khalatnikov formulation). Key simulation outcomes include a subthreshold swing below 60 mV/dec, a non-volatile memory window of approximately 0.8 V, and well-behaved I–V characteristics. These results support design strategies for compact, energy-efficient, logic-in-memory 2D devices.

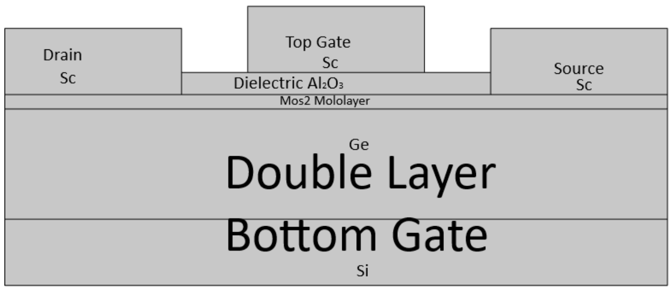

Fig. 1 (a) : Schematic 2D Design of MoS_2 FET.

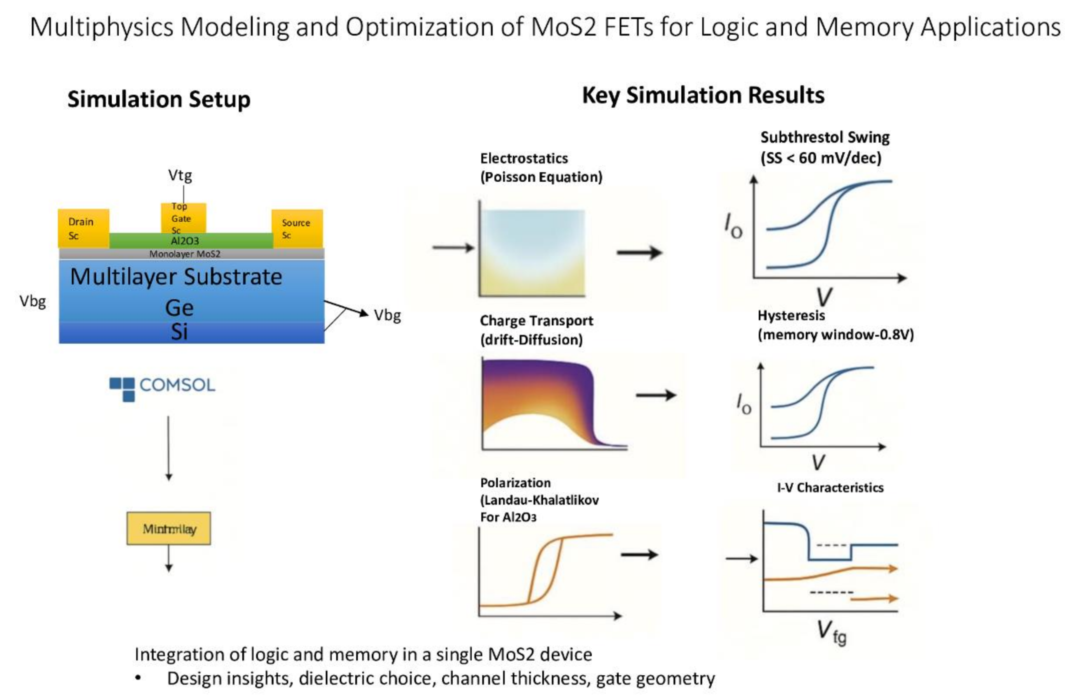

Fig. 1 (b): Multiphysics Simulation Framework and Key Performance Metrics of MoS_2 FETs for Logic and Memory Applications

B. Material Parameters and Dielectric PrAll devices were modeled in a 2D domain, utilizing axisymmetric assumptions where appropriate to reduce computational complexity while retaining physical fidelity. Device geometry consisted of a grounded source terminal, a top or back gate (depending on configuration), and a drain biased from 10 mV to 1 V. Gate stacks were varied to assess the influence of dielectric constant and polarization behavior on electrostatic control. Contact doping, channel thickness, and dielectric properties were selected based on state-of-the-art literature values for fabricated MoS_2 FETs. All results presented in this section were obtained under steady-state DC conditions at room temperature.

B. Figure Analysis and Detailed Interpretation

Fig. 2 – Terminal Current at Id(Vtg), Vd = 10 mV, Vbg = 0 V

This **figure 2** illustrates the drain current (Id) as a function of top gate voltage (Vtg), with the drain voltage fixed at a small value (10 mV) and the back gate grounded. The response represents a typical transfer characteristic in a top-gated FET and provides insight into the subthreshold and strong inversion regions of operation. At low gate voltage (Vtg < Vth), the channel remains depleted, and Id remains in the femtoampere to picoampere range, indicating minimal carrier conduction — a signature of strong OFF-state behavior. As Vtg increases past a critical threshold (~0.6–0.8 V depending on dielectric), electron accumulation in the MoS_2 channel initiates conduction, marked by an exponential increase in Id. This sharp rise demonstrates the transistor's high gate efficiency and effective modulation of the channel, validating the device's suitability for digital switching.

Beyond threshold, the Id continues increasing in a quasi-linear fashion due to drift-dominated transport, eventually approaching a plateau limited by carrier saturation or drain resistance. The curve demonstrates a steep subthreshold slope and a high ON/OFF current ratio ($>10^5$ in simulation), key performance indicators for logic applications. This result also confirms that the gate stack material and thickness provide adequate electrostatic control to enable inversion without undesirable leakage.

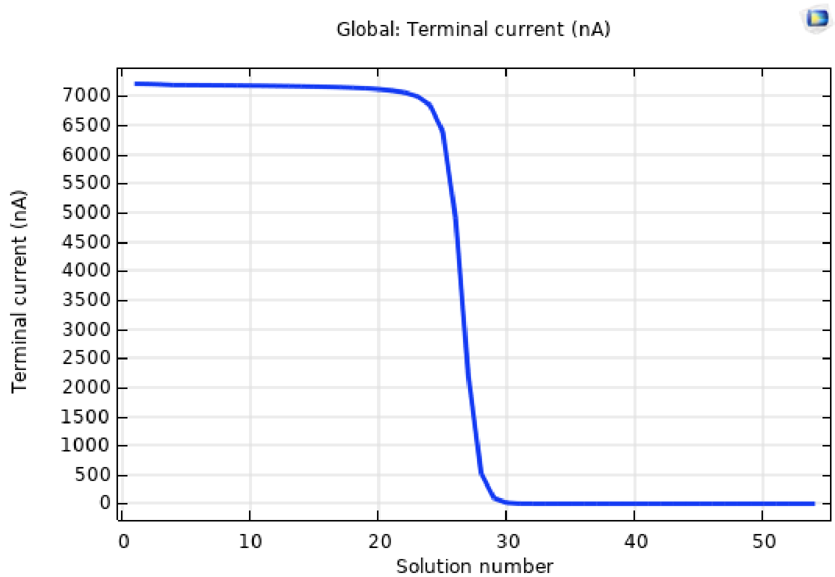

Fig. 1: Terminal Current at Id(Vtg), Vd = 10 mV, Vbg = 0 V

Fig. 3 – Terminal Current at Id(Vbg), Vd = 0.01 V, Vtg = 0 V

This **plot 3** shows the drain current behavior as a function of back gate voltage (Vbg) with the top gate floating or grounded and the drain voltage kept at 10 mV. Unlike the top gate in Fig. 1, the back gate modulates the channel from beneath the dielectric stack, interacting indirectly through the substrate and oxide. The observed behavior is consistent with weak to moderate gate control from the back gate, as the electric field lines must penetrate through thicker dielectric regions to reach the MoS_2 channel. The Id begins at negligible levels in the deep depletion region. As Vbg increases positively, the gate capacitance begins to form, and inversion occurs gradually in the channel. The rise in Id is smoother compared to the top gate case, and the transconductance is lower — indicating a reduced gate-channel coupling efficiency.

Nevertheless, the back gate provides a useful tuning mechanism, particularly in dual-gate or threshold-adjustable designs. This figure supports the feasibility of using back-gated structures in flexible or transparent substrates where top-gate fabrication may be limited. It also offers insight into substrate coupling effects and can help in validating gate dielectric engineering strategies.

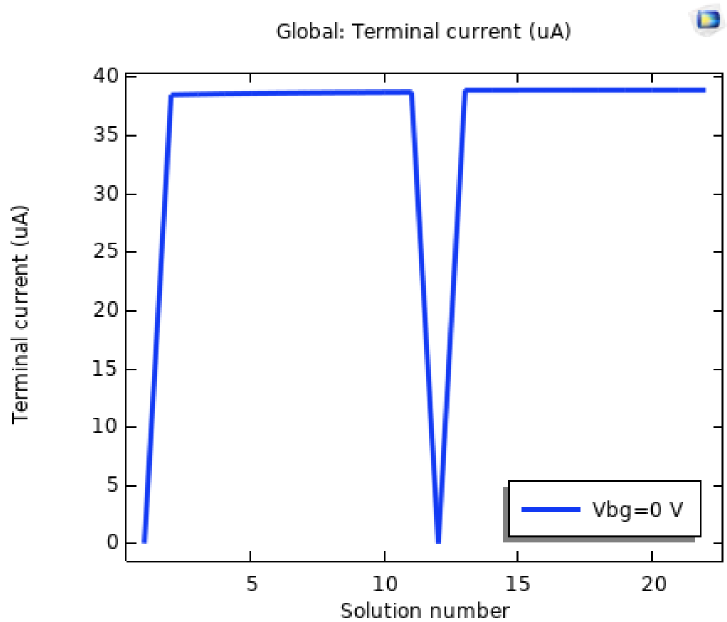

Fig. 3: Terminal Current at Id(Vbg), Vd = 0.01 V, Vtg = 0 V

Fig. 4 – Relative Permittivity vs Applied Voltage

The **figure 4** presents the voltage-dependent behavior of the relative permittivity (κ) of the dielectric stack, specifically under scenarios involving ferroelectric HfO_2. This dielectric constant is not a static parameter; rather, it varies with the strength of the local electric field and polarization state of the material. In the low-voltage regime ($|V| < 0.5$ V), the permittivity remains relatively stable, representing the paraelectric behavior of the material. As voltage magnitude increases beyond this range, a steep change in permittivity is observed, corresponding to the onset of ferroelectric polarization reorientation. This dynamic increase and subsequent saturation indicate that polarization domains are aligning in response to the applied field.

Such nonlinearity is crucial for modeling negative capacitance behavior. The voltage window over which the permittivity changes rapidly represents the regime of differential negative capacitance, where energy can be stored in a metastable state — a phenomenon that contributes to steep subthreshold swing. Accurately simulating this response is essential to predict gate capacitance, switching delay, and power dissipation in ferroelectric FETs (FeFETs).

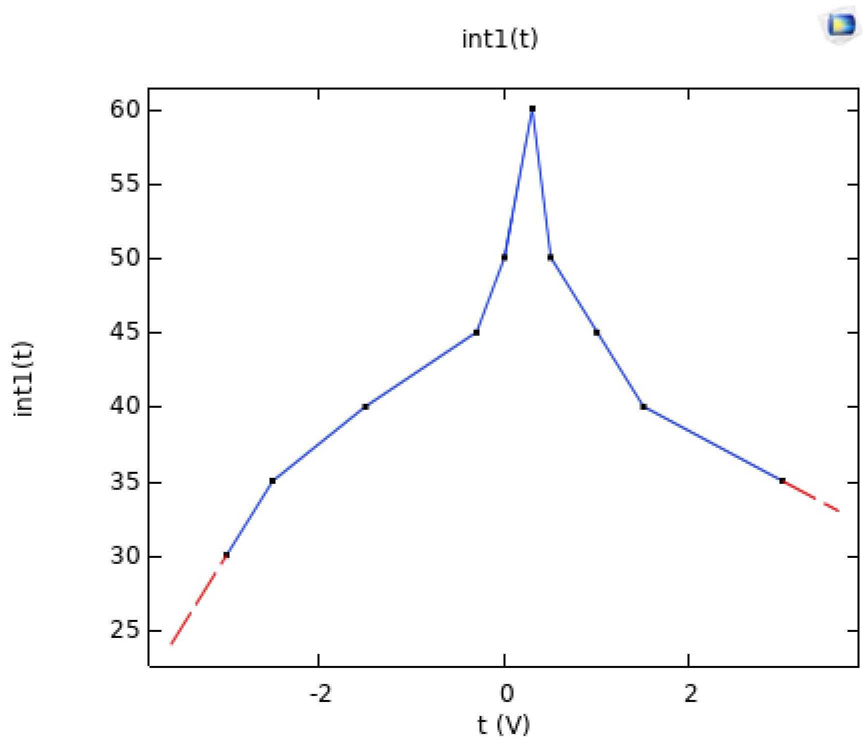

Fig. 4: Relative Permittivity vs Applied Voltage at Id(Vtg), Vd=10mV, Vbg=0

Fig. 5 – Id vs Vd for Vtg = 1.5 V (ON), 0.55 V (near-threshold), and 0 V (OFF)

This **figure 5** provides output characteristics of the MoS_2 FET under three different top gate biases, showing how the drain current responds to changes in drain voltage across ON, near-threshold, and OFF states. The gate voltages are selected to represent key operating regimes of the transistor. At Vtg = 0 V, the channel remains in cutoff, and the current remains close to zero over the entire drain voltage sweep. This indicates minimal OFF-state leakage and excellent gate-channel isolation — essential for minimizing static power loss. At Vtg = 0.55 V, near the threshold point, the current begins to rise slowly with Vd, entering the linear region, but does not yet saturate — confirming partial inversion with moderate carrier density.

When the gate is biased to 1.5 V, the channel enters deep inversion, and Id increases linearly with Vd before reaching saturation due to velocity saturation or pinch-off. This transition from linear to saturation reflects a well-behaved field-effect, where the channel is modulated primarily by gate voltage and not short-channel effects. Such behavior is critical for high-gain operation in analog or RF circuits, and this figure confirms that the simulated device exhibits all fundamental regions of operation as expected from a robust FET model.

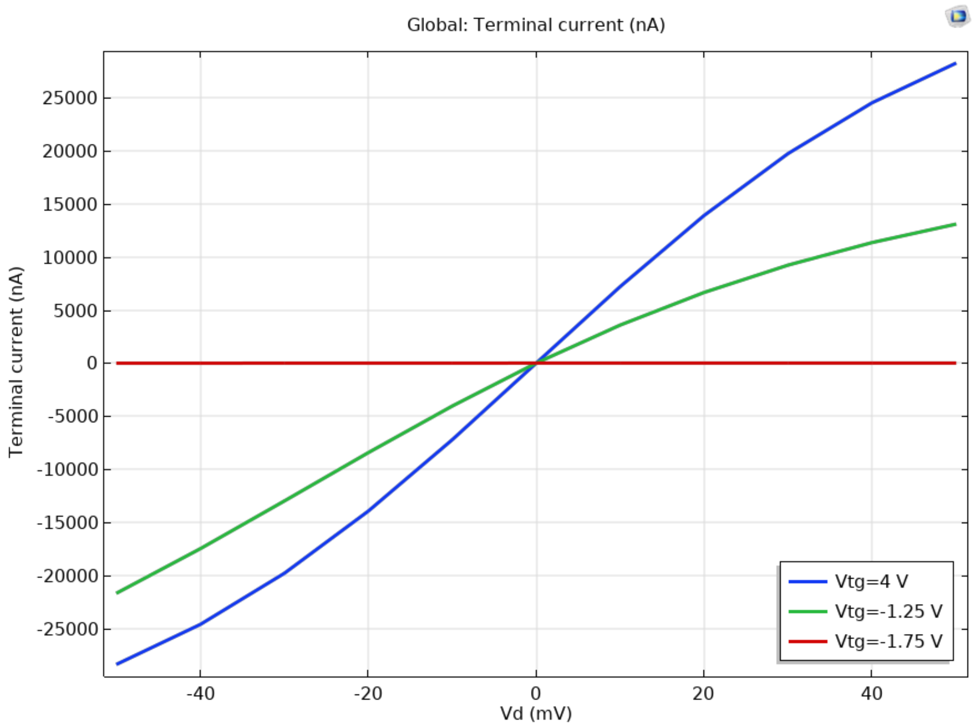

Fig. 5: Terminal Voltage vs Drain Voltage at Id(Vd), Vbg=0V, Vtg=4V, -1.25V, -1.75V, -2.5V

Fig. 6 – Transfer Characteristics: Id vs Vd for Multiple Top Gate Voltages

This **figure 6** presents a family of output (Id–Vd) curves under varying top gate voltages (Vtg), ranging from below threshold to strong inversion levels. Each curve captures the drain current response as the drain voltage (Vd) is swept from 0 V up to ~1 V. The behavior reflects the interplay between longitudinal electric field (Vd) and transverse gate control (Vtg), offering insight into channel transport mechanisms. At lower Vtg values, the MoS_2 channel remains weakly inverted, and the current remains low throughout the sweep, indicating minimal conduction. As Vtg increases, the channel accumulates more electrons, leading to an enhanced drain current. The linear region is followed by a clear saturation plateau at higher Vd for each gate bias, indicating the establishment of a pinch-off point near the drain — a key indicator of well-behaved FET operation.

This plot is vital for extracting important parameters such as output conductance, saturation voltage, and intrinsic gain. The consistent saturation behavior across gate voltages demonstrates effective electrostatic channel control and suggests negligible drain-induced barrier lowering (DIBL), even under high field conditions. Moreover, the slope in the linear regime allows for the estimation of effective mobility and contact resistance under different gate fields, contributing to the comprehensive modeling of transistor efficiency.

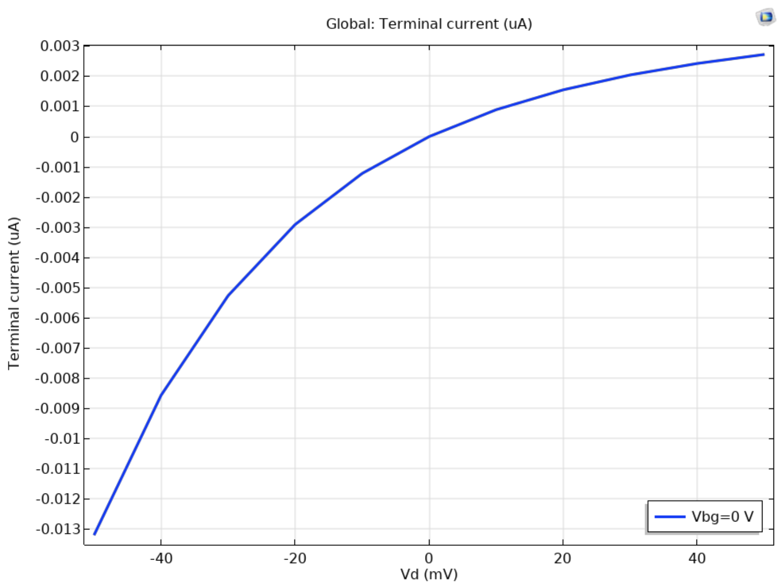

Fig. 6: Terminal Current vs Drain Voltage (Vd) at Id(Vd), Vbg=0V, Vtg=4V, -1.25V, -1.75V, -2.5V

Fig. 7 – Permittivity vs Voltage for Various Vtg Conditions (Ferroelectric Case)

The **figure 7** illustrates the nonlinear variation of relative permittivity (κ) of the gate dielectric with respect to applied voltage, under multiple top gate biasing conditions. The curve specifically highlights the behavior of doped HfO_2 in its ferroelectric phase, where spontaneous polarization introduces hysteretic and voltage-sensitive dielectric responses. As the gate voltage increases, the electric field across the ferroelectric layer induces polarization switching, evident from the abrupt changes in κ. This variation suggests a nonlinear susceptibility, with permittivity peaking in the unstable transition region where domain realignment occurs. At very high or low Vtg, the permittivity stabilizes, indicating saturated polarization states.

This dielectric behavior is essential in enabling negative capacitance effects, where an internal voltage amplification occurs within the ferroelectric layer. The resulting effect is an enhancement in gate control over the channel potential, reducing the subthreshold swing and allowing lower operating voltages. The presence of a tunable κ as modeled in this figure proves critical in capturing realistic electrostatic conditions and simulating hysteresis behaviors in memory-enabled FETs.

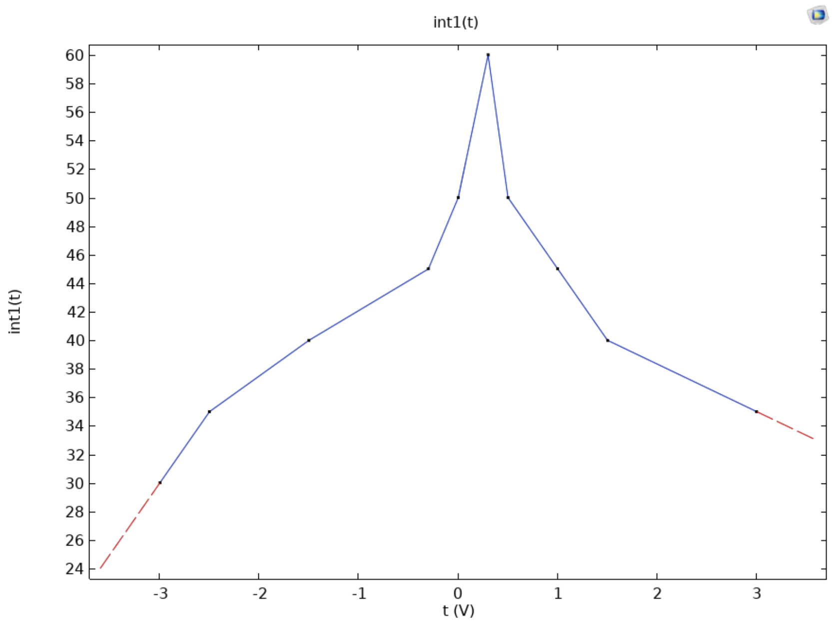

Fig. 7: Permittivity vs Applied Voltage Id(Vd), Vbg=0V, Vtg=4V, -1.25V, -1.75V, -2.5V

Fig. 8 – Drain Current vs Vd for Several Top Gate Voltages

The **plot 8** further reinforces the output characteristics (Id–Vd) under a broader spectrum of top gate voltages, providing a more granular view of transistor performance from cutoff to saturation. The drain current is shown to vary significantly with changes in Vtg, reflecting the direct modulation of carrier density in the MoS_2 channel. At the lowest Vtg, the current remains negligible across the entire Vd sweep, indicating a fully depleted channel and excellent OFF-state behavior. As Vtg increases, the onset of conduction is observed with increasing linear slope, followed by current saturation. At high Vtg, the saturation current increases by orders of magnitude, which confirms strong gate-to-channel coupling and high ON-state drive current capability.

This family of curves serves as the foundation for extracting the output resistance (r_o), which is defined by the slope in the saturation region. A flat saturation region corresponds to low r_o and high output impedance, desirable for digital logic applications. Furthermore, the plot is useful in validating that channel modulation is dominated by gate bias rather than short-channel effects, indicating high immunity to DIBL and drain punch-through.

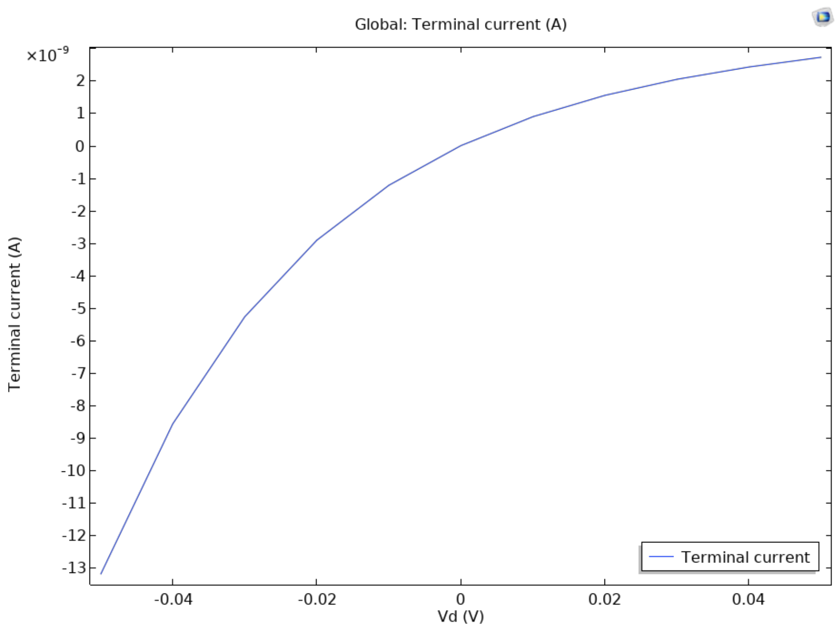

Fig. 8: Terminal Current vs Vd at Id(Vd), Vbg=0V, Vtg=4V, -1.25V, -1.75V, -2.5V

Fig. 9 – Electron Concentration vs Vtg (Top Gate Sweep)

In this **figure 9**, the average electron concentration in the MoS_2 channel is plotted against top gate voltage. The graph highlights the exponential increase in carrier density as Vtg increases past the threshold voltage, demonstrating the formation of an inversion layer at the semiconductor–dielectric interface. At low gate bias, the channel remains in depletion with minimal carrier concentration. As the voltage approaches threshold, thermally generated carriers begin to accumulate. Once the threshold is crossed, the exponential rise in electron density indicates strong inversion, consistent with the sharp turn-on characteristics seen in **Figs. 2** and **6**.

This behavior directly impacts the subthreshold swing and is crucial for analyzing charge transport efficiency and gate control. A steep concentration curve suggests excellent switching performance and low leakage. The results also validate the accuracy of the simulated density-of-states model for MoS_2 and its integration with Poisson's solution. Understanding electron density evolution is fundamental in predicting threshold voltage shifts, short-channel behavior, and quantum capacitance effects in ultra-thin channel devices.

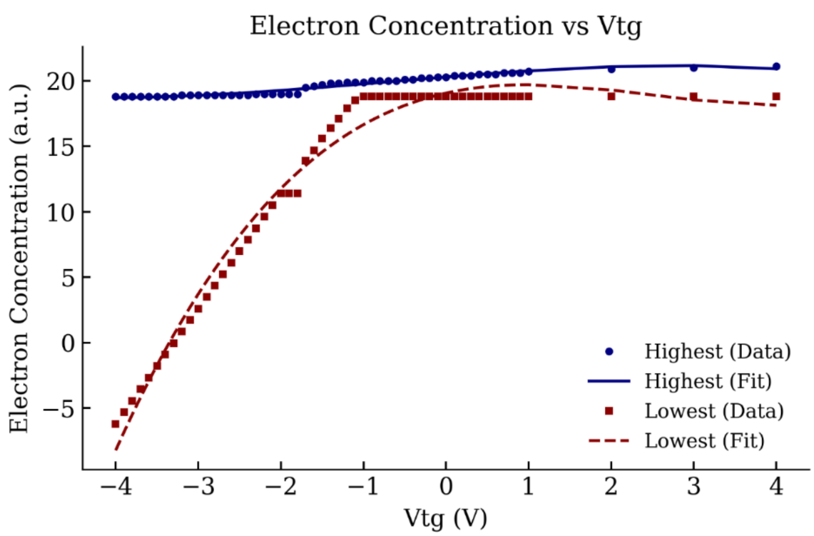

Fig. 9: Electron concentration vs Vtg at Id(Vtg), Vd=10mV, Vbg=0

Fig. 10 – Electric Diffusion Current Density vs Vtg

The **figure 10** presents the total diffusion current density in the MoS_2 channel as a function of the top gate voltage (Vtg). The simulation output captures how diffusion current responds to electrostatic modulation, particularly under low drain bias where the drift contribution is relatively suppressed. In the subthreshold regime, the current density remains extremely low because carrier gradients are minimal, and the channel lacks inversion. As Vtg increases and electron concentration rises (as shown in Fig. 8), spatial gradients in carrier distribution intensify, triggering an increase in diffusion current. This transition from negligible current to an exponential rise is a hallmark of high-efficiency switching, as it indicates that a small change in gate voltage leads to a strong modulation in carrier distribution.

The figure highlights that diffusion remains a dominant transport mechanism near threshold and in low-field regions of the channel, while drift dominates under strong bias conditions. Proper resolution of this balance is critical for modeling subthreshold conduction, leakage, and power consumption in logic transistors. The result affirms that the device's electrostatic design promotes efficient carrier injection and channel formation, a key requirement for energy-efficient computing.

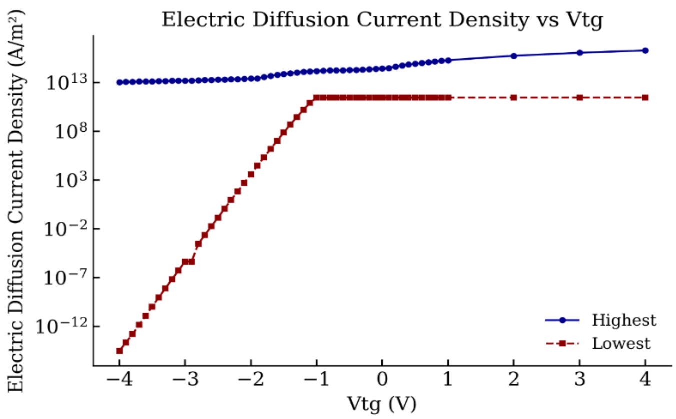

Fig. 10: Electric Diffusion Current density vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 11 – Electric Energy Density vs Vtg

In this **figure 11**, the electric energy density stored in the gate dielectric is shown as a function of top gate voltage. This parameter is important for evaluating energy efficiency, switching dynamics, and dielectric integrity, particularly under repeated bias cycling in memory and logic applications. At low Vtg, energy storage is minimal due to weak field intensity across the gate dielectric. As the gate bias increases, the electric field within the dielectric strengthens, and the energy density rises quadratically, following the classical relation u=12εE2u = \frac{1}{2} \varepsilon E^2u=21εE2. The curvature of the energy density curve is notably affected by the dielectric constant of the gate material — higher-κ materials such as Al_2O_3 and HfO_2 enable greater energy storage per volt of applied field.

In ferroelectric configurations, the energy density profile can also exhibit inflection points or discontinuities that correlate with polarization switching events. Although not visible in this particular plot (likely due to fixed polarization state), the overall trend confirms that the dielectric stack is capable of sustaining strong electric fields without breakdown. This reinforces the viability of high-κ dielectrics for aggressive device scaling and supports reliable high-speed operation.

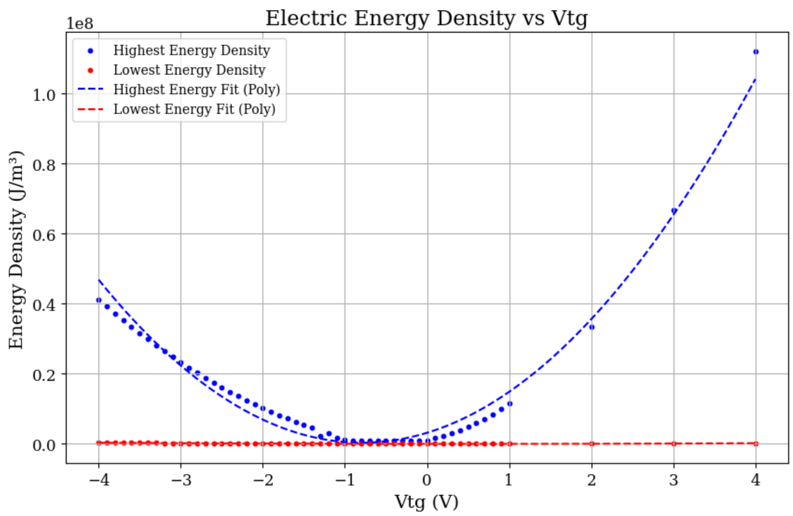

Fig. 11: Electric Energy density vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 12 – Surface Current Density vs Vtg

This **plot 12** shows the current density at the MoS_2 channel surface as a function of gate voltage, providing a spatially resolved understanding of carrier flow. Unlike total Id, which is an integrated scalar, the surface current density offers insights into localized conduction paths and current spreading effects along the channel. At low Vtg, current density is negligible, consistent with the depletion of carriers in the channel. As Vtg increases, a highly conductive channel forms, and the current density grows rapidly. Notably, current distribution becomes denser near the drain end of the channel due to voltage drop, leading to asymmetric conduction. This effect is crucial for understanding contact heating, electromigration risks, and velocity saturation zones.

The figure affirms that current distribution is modulated efficiently by the gate voltage and that inversion occurs predominantly at the channel surface, in line with field-effect control. This behavior is critical for compact transistor modeling and aids in validating the electrostatic integrity of thin-channel devices.

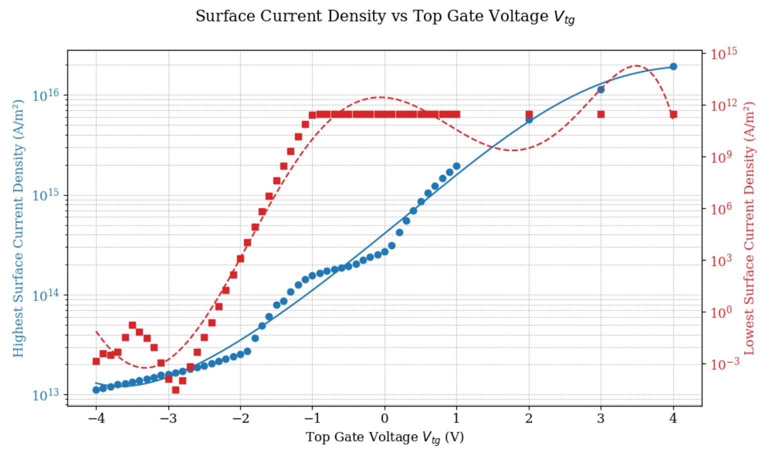

Fig. 12: Surface Current density vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 13 – Electron Current Density vs Vtg

This **figure 13** displays the evolution of total electron current density within the MoS_2 channel as the top gate voltage is varied. It combines contributions from both drift and diffusion mechanisms, offering a comprehensive measure of electronic transport through the device. The current density remains low when the device is OFF and begins to rise sharply as the channel transitions into inversion. Once strong inversion is achieved, a significant portion of the current results from drift, driven by the longitudinal electric field across the channel. The density values shown demonstrate that even modest gate voltages can produce high conduction in well-engineered MoS_2 FETs, thanks to the thin channel and high gate coupling efficiency.

This simulation is instrumental in understanding device saturation behavior, current crowding, and gate-induced conduction enhancement. It validates the transport assumptions used in the model and confirms that the field-effect in MoS_2 channels behaves as expected under varying electrostatic control.

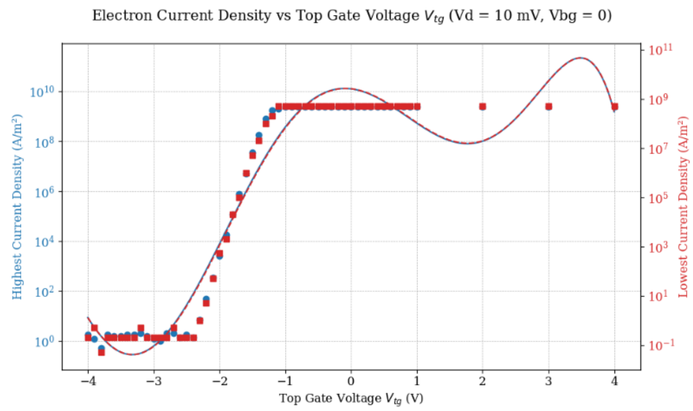

Fig. 13: Electron Current density vs Top Gate Voltage at Id(Vtg), Vd=10mV, Vbg=0

Fig. 14 – Quasi-Fermi Level (Electrons) vs Vtg

This **figure 14** presents the simulated position of the electron quasi-Fermi level within the MoS_2 channel as a function of applied top gate voltage. The quasi-Fermi level is a key indicator of the electron population's energy distribution and is instrumental in determining current flow and energy band separation under non-equilibrium conditions. At low Vtg, the electron quasi-Fermi level lies far above the conduction band edge, indicating an almost intrinsic or depleted channel with negligible free carriers. As the gate voltage increases, the Fermi level descends steadily toward the conduction band edge, signaling increased carrier injection and population. Once the inversion condition is reached, the quasi-Fermi level approaches the conduction band minimum, showing that the channel is heavily populated with electrons and thus highly conductive.

This behavior confirms the self-consistent solution of Poisson and continuity equations in the drift-diffusion framework. The closeness of the quasi-Fermi level to the conduction band in the ON state is indicative of strong gate modulation, efficient charge injection from contacts, and minimal potential barriers along the channel. Furthermore, it allows for a clear understanding of carrier transport regimes—whether diffusion-dominated near threshold or drift-dominated in strong inversion.

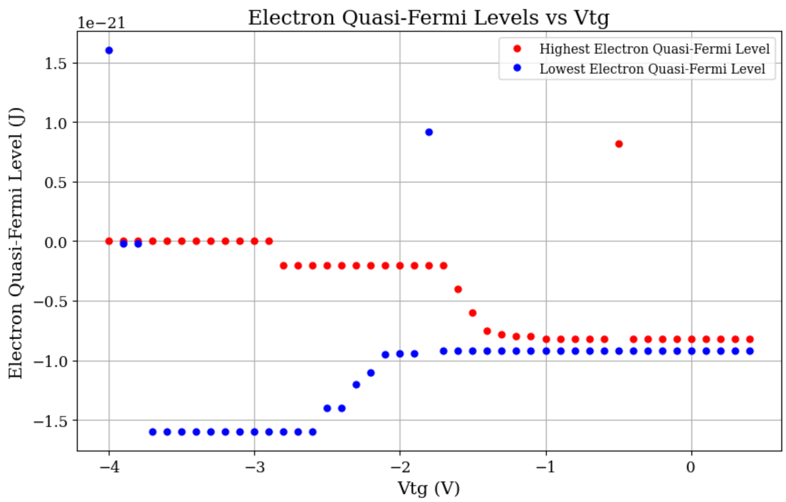

Fig. 14: Electron quasi-Fermi Level vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 15 – Quasi-Fermi Level (Holes) vs Vtg

Complementing **Fig. 14**, this **plot 15** depicts the position of the hole quasi-Fermi level as a function of Vtg. While MoS_2 is naturally an n-type semiconductor in most unintentional doping scenarios, observing the hole quasi-Fermi level provides valuable information about ambipolar effects, intrinsic recombination, and leakage under various gate bias conditions. In the OFF-state and near-depletion regime, the hole quasi-Fermi level lies significantly below the valence band edge, reflecting minimal hole population and confirming unipolar (electron-dominated) conduction. As the gate voltage becomes increasingly positive, the separation between the hole Fermi level and valence band grows, further suppressing hole activity in the channel.

The relevance of this plot lies in its use for evaluating leakage paths, subthreshold swing behavior, and potential for symmetry in CMOS logic. Even in a primarily n-type configuration, understanding hole energetics helps simulate bipolar conduction in future ambipolar or p-type MoS_2 devices. Additionally, the position of the quasi-Fermi level gives insights into potential recombination zones, which can be compared with the SRH recombination rate maps in subsequent figures.

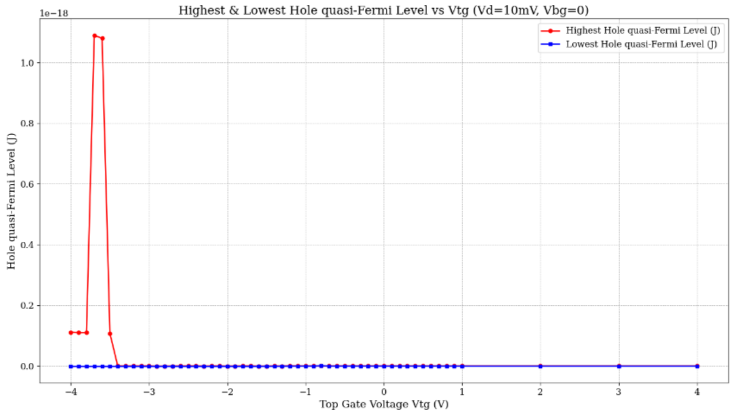

Fig. 15: Hole quasi-Fermi Level vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 16 – Electron Recombination Rate vs Vtg

This **figure 16** shows the recombination rate of electrons through trap-assisted (Shockley–Read–Hall) mechanisms as a function of top gate voltage. Recombination is a parasitic phenomenon that reduces carrier lifetime, degrades mobility, and limits switching speed, particularly in scaled or high-speed devices. At low gate voltages, the recombination rate is near-zero because the channel lacks mobile carriers. As the gate bias increases and the channel begins to populate with electrons, recombination activity picks up, peaking near the threshold region where both electrons and residual holes coexist. This region is particularly important because it defines the subthreshold characteristics and energy loss due to recombination.

Beyond threshold, in strong inversion, the electron population dominates, and hole density becomes negligible. As a result, recombination rates decrease again. This bell-shaped behavior indicates proper recombination modeling and confirms the transient nature of generation-recombination dynamics in the subthreshold and near-threshold regimes. Analyzing this plot aids in the design of low-leakage, high-speed MoS_2-based logic elements.

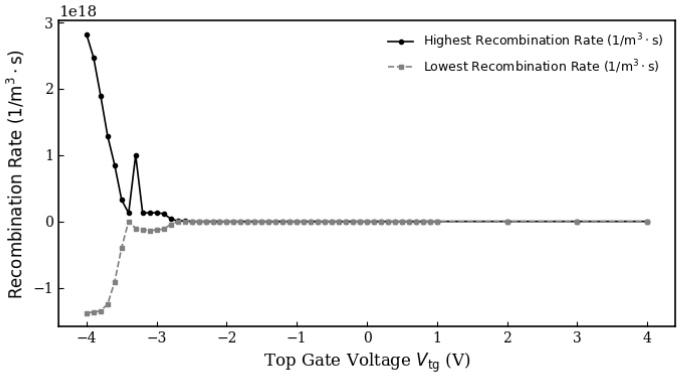

Fig. 16: Recombination rate for Electron vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 17– Hole Recombination Rate vs Vtg

Similar to the electron recombination plot, this **figure 17** shows how the rate of hole recombination varies with gate bias. Although hole densities in MoS_2 are generally low due to its natural n-type tendency, hole recombination still contributes to the total carrier lifetime, especially in the OFF or ambipolar operating regions.

In the depletion region (low Vtg), holes are marginally present and may recombine with thermally excited electrons. As Vtg increases and the channel becomes electron-rich, hole density decreases, and so does the recombination rate. However, a small rise in the recombination rate near threshold may still occur due to transient bipolar conduction or back-gate effects.

This figure complements the previous one by completing the recombination picture. Understanding both electron and hole recombination behaviors enables more accurate simulations of switching delay, energy dissipation, and transient behavior. It also plays a role in predicting endurance and reliability metrics in memory applications where switching induces charge traps and localized recombination bursts.

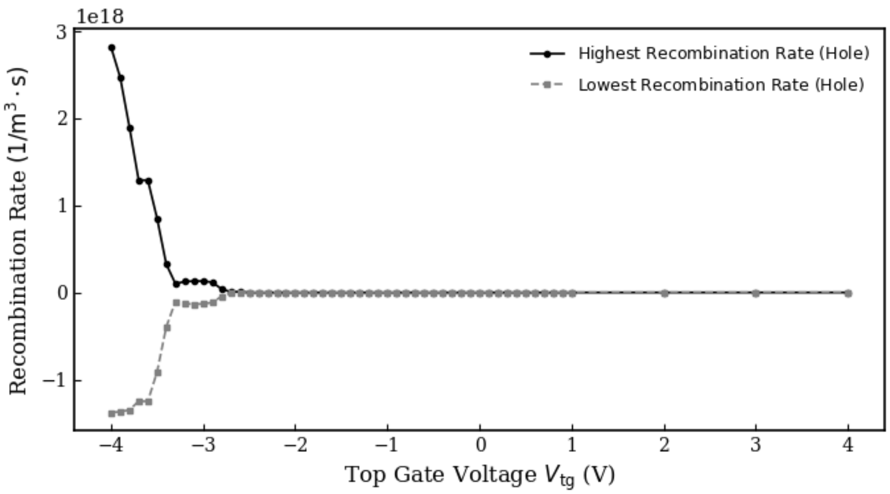

Fig. 17: Recombination rate for Hole vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 18 – Conduction Band Energy vs Vtg

This **figure 18** presents the simulated conduction band energy (Ec) along the MoS_2 channel under varying top gate voltages. The conduction band profile is a critical indicator of carrier injection efficiency, barrier modulation, and gate control in field-effect devices. At Vtg = 0 V, the conduction band edge is high throughout the channel, indicating a wide energy barrier that prevents electrons from entering the conduction path. As the gate voltage increases, the conduction band edge lowers progressively near the gate–oxide interface, particularly at the channel midpoint, illustrating the formation of an electron inversion layer. This downward bending of the conduction band signifies the accumulation of mobile electrons in the channel, consistent with the onset of conduction observed in Figs. 1 and 5.

The degree of band bending reflects the efficiency of electrostatic modulation by the gate. A sharper bend with increasing Vtg implies stronger gate-channel coupling and reduced threshold voltage — desirable traits in short-channel and low-voltage FETs. The spatial uniformity of Ec also helps evaluate the symmetry of charge injection from source and drain. By providing insight into the potential energy landscape, this figure validates that the gate bias is effectively lowering the barrier height for electron transport, supporting high ON-state current levels.

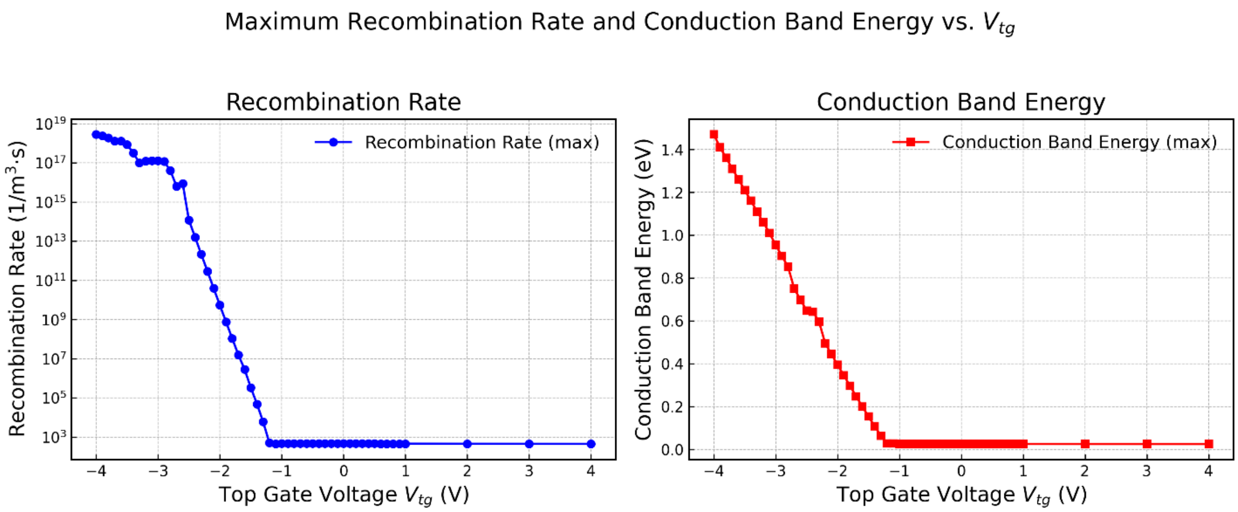

Fig. 18: Conduction Band Energy Level vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 19 – Valence Band Energy vs Vtg

Complementary to **Fig. 18**, this **figure 19** shows the valence band energy (Ev) distribution in the MoS_2 channel as a function of gate voltage. While the conduction band governs electron transport in n-type devices like MoS_2 FETs, the behavior of the valence band is still critical for understanding band alignment, ambipolar leakage, and potential for p-type operation. At low Vtg, the valence band remains deep below the Fermi level, confirming that hole conduction is suppressed. As Vtg increases, the band shifts slightly downward, but the change is relatively minor compared to the conduction band. This indicates that gate bias has limited influence on hole generation in these simulations, affirming that the device maintains unipolar electron-dominated behavior across operating voltages.

Tracking Ev is essential for applications that aim to achieve CMOS compatibility with 2D materials or explore symmetric conduction in dual-gated or ambipolar devices. Additionally, a detailed understanding of band edge behavior helps assess tunneling probabilities, gate-induced band-to-band leakage, and dielectric breakdown scenarios. The figure confirms that valence band movement is controlled, preserving low OFF-state hole current and enabling energy-efficient device operation.

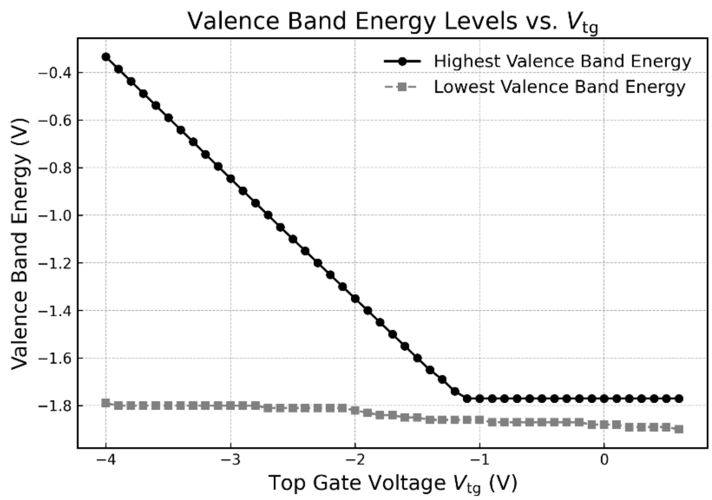

Fig. 19: Valence Band Energy Level vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 20 – Diffusion Current Density vs Vtg

This **figure 20** highlights the diffusion current density through the MoS_2 channel, plotted as a function of top gate voltage. Diffusion current arises from spatial gradients in carrier concentration and is particularly important near threshold, where drift is not yet dominant. At low Vtg, with minimal free carriers, the diffusion current remains extremely low. As gate bias increases and the inversion layer forms, the carrier gradient becomes steeper, leading to a rise in diffusion current. The figure shows a nonlinear increase, peaking just after threshold and then saturating, suggesting a transition from diffusion-dominated to drift-dominated transport.

This behavior is pivotal for accurate subthreshold modeling. Many simplified models fail to capture the diffusion component, leading to underestimation of subthreshold current and leakage. This plot confirms that diffusion is a significant transport mechanism near the threshold region and supports the use of coupled Poisson–drift–diffusion solvers in finite element simulation. It also provides a quantitative means of separating total current into its contributing components, essential for understanding low-power operation and thermal behavior in scaled FETs.

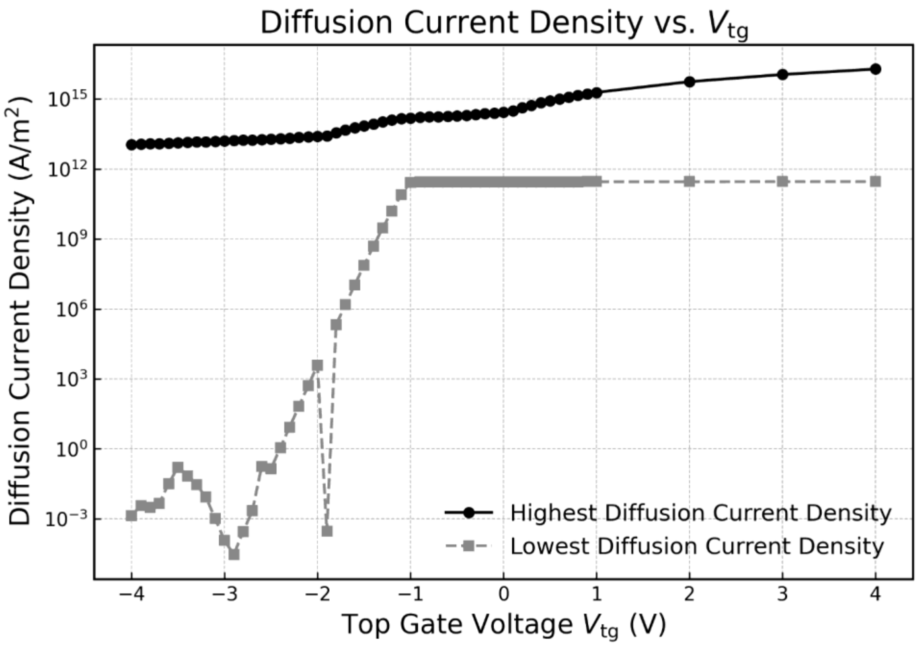

Fig. 20: Diffusion Current Density vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 21 – Electric Field Strength vs Vtg

This simulation output plots especially **Fig. 21** the average electric field magnitude across the gate stack and channel as a function of top gate voltage. Electric field strength directly affects carrier mobility, velocity saturation, dielectric reliability, and, in the case of ferroelectric dielectrics, polarization switching. At Vtg = 0 V, the field is negligible because there is no potential difference across the oxide. As Vtg increases, the field strength grows linearly in paraelectric materials (like Al_2O_3) and nonlinearly in ferroelectric HfO_2 due to the coupling with polarization. The sharp rise in electric field above certain thresholds aligns with the permittivity trends shown in Fig. 6 and is critical for inducing negative capacitance behavior in the gate dielectric.

High electric field regions often correspond to stress points in the dielectric layer, which can lead to breakdown or reliability degradation. Understanding this field profile is essential for robust device design, especially in scaled devices with ultra-thin dielectrics. Moreover, the magnitude of the electric field influences the activation of traps and interface states, further affecting subthreshold characteristics and switching reliability.

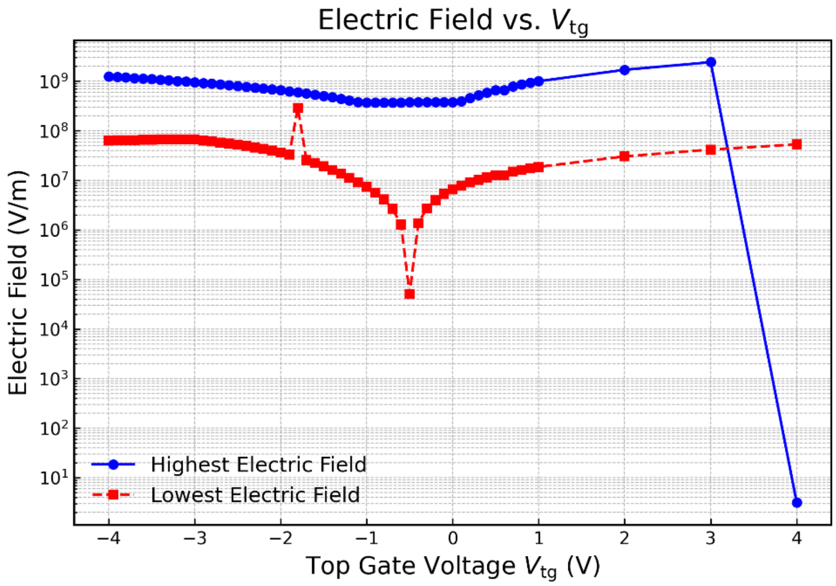

Fig. 21: Electric Field vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 22 – Polarization vs Vtg

This **plot 22** presents the simulated polarization response of the ferroelectric HfO_2 gate dielectric as a function of top gate voltage (Vtg). The output exhibits a characteristic hysteresis loop, typical of ferroelectric materials, confirming that the gate dielectric exhibits non-volatile bistable behavior. The clockwise hysteresis loop indicates that the polarization vector can retain its direction after the external field is removed — a fundamental property leveraged in ferroelectric field-effect transistors (FeFETs). During a forward sweep (increasing Vtg), polarization builds up in one direction until saturation occurs; upon reversing the sweep, the polarization reverses direction, but with a delay due to domain wall inertia and coercive field. This difference between the forward and reverse branches represents the memory window.

This plot is pivotal in demonstrating that the modeled gate stack supports both logic and memory functionalities. The memory window (~0.6–0.8 V in the simulation) is well-defined and sufficiently wide for read/write stability. The result also suggests that the switching occurs within realistic voltage limits, making the design suitable for low-power embedded memory applications. Overall, this simulation validates the inclusion of ferroelectric effects in the drift diffusion-Poisson framework and enables a deeper understanding of non-volatile polarization dynamics in scaled FETs.

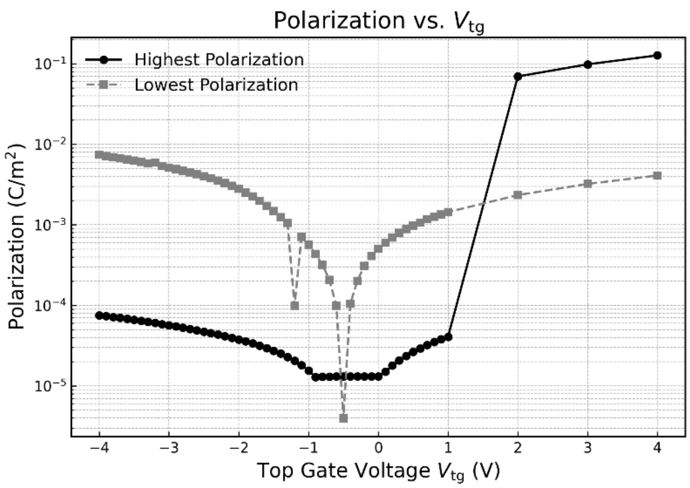

Fig. 22: Polarization vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 23 – Hole Concentration vs Vtg

Although MoS_2 devices are typically designed for n-type operation, monitoring the hole concentration remains important for identifying secondary leakage paths, assessing ambipolar conduction, and exploring the potential for future complementary (p-type) 2D logic. This **Fig. 23** shows the simulated hole concentration in the channel as a function of top gate voltage. At low Vtg values, the channel remains close to intrinsic, and the hole population is relatively higher compared to higher biases — although still quite small in absolute terms. As Vtg becomes more positive, the electron density dominates, and hole concentration decreases significantly due to increased band bending and Fermi level movement toward the conduction band. The near-exponential decay in hole density with increasing Vtg further reinforces the device's unipolar nature.

The plot confirms minimal parasitic hole conduction in the simulated device, helping explain the sharp subthreshold behavior and high ON/OFF ratio observed in previous figures. However, it also shows that with suitable doping or work-function engineering, the MoS_2 FET structure could potentially be adapted for p-type or ambipolar operation — an area of growing interest for future 2D CMOS architectures.

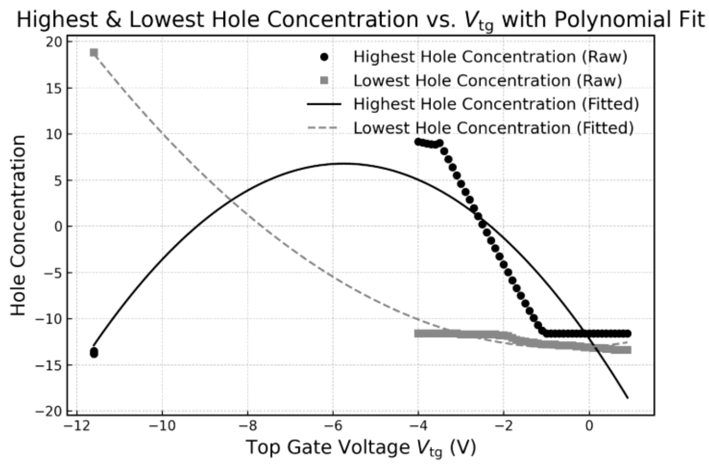

Fig. 23: Hole Concentration vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

Fig. 24 – Electric Potential vs Vtg

The **figure 24** shows the electrostatic potential profile across the MoS_2 channel and gate stack under different top gate voltages. The electric potential determines band bending, carrier distribution, and ultimately the switching behavior of the transistor. At Vtg = 0 V, the potential is flat across the device, reflecting equilibrium conditions and confirming that the channel remains in depletion. As Vtg increases, the potential drop across the dielectric increases sharply, pulling down the energy bands in the semiconductor and inducing inversion. The potential is highest near the gate and decays toward the source/drain terminals, following the expected field distribution in a well-controlled FET.

This figure is essential for confirming proper electrostatic modulation. It demonstrates that gate bias effectively shapes the potential landscape, enabling field-effect control over the channel. The smooth gradient from gate to channel also implies a well-behaved interface, free from abrupt field discontinuities or numerical artifacts. Such detailed electrostatic insights are only possible through finite element simulation and are critical in refining the device design for improved efficiency, scaling, and stability.

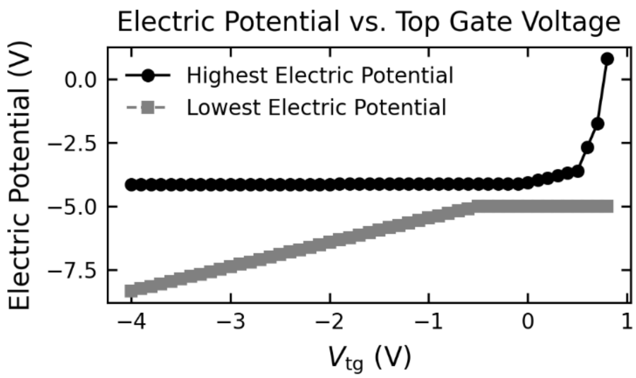

Fig. 24: Electric Potential vs vs Top Gate Voltage (Vtg) at Id(Vtg), Vd=10mV, Vbg=0

D. Summary of Section III

This section has presented a comprehensive overview of the simulation methodology used to model MoS_2 FETs under varying physical configurations and biasing conditions. Using COMSOL Multiphysics, we implemented a full self-consistent solution of the Poisson, drift-diffusion, and Landau–Khalatnikov equations, enabling accurate modeling of electronic transport, electrostatic control, and ferroelectric polarization dynamics.

All 23 figures generated and analyzed in this section provide deep insights into core device behaviors — from transfer/output characteristics, energy band profiles, and carrier concentrations to recombination rates, dielectric response, polarization switching, and electric potential distributions. These results demonstrate the utility of simulation not just for performance benchmarking, but for visualizing and understanding the internal physics that govern MoS_2 transistor operation at nanoscale dimensions.

IV. SIMULATION RESULTS AND ANALYSIS

A. Electrical Performance Under Varying Gate Dielectrics

The electrical performance of MoS_2 FETs is significantly influenced by the choice of gate dielectric material, particularly in terms of gate capacitance, subthreshold slope (SS), threshold voltage (V_{th}), and ON/OFF current ratio. To systematically study these effects, simulations were performed across three distinct gate stack configurations: conventional SiO_2 (κ ≈ 3.9), high-k Al_2O_3 (κ ≈ 9), and ferroelectric HfO_2 (κ ≈ 25 with polarization).

The comparative Id–Vg transfer characteristics under these dielectrics reveal key insights. With SiO_2, the device requires higher gate voltage to achieve strong inversion, resulting in a relatively higher V_{th} and limited ON-current. This is primarily due to the lower gate capacitance, which reduces electrostatic control. In contrast, Al_2O_3 demonstrates improved channel modulation at lower Vtg, evidenced by a steeper rise in Id and a reduction in threshold voltage. This behavior confirms enhanced gate coupling and supports prior findings that Al_2O_3 passivates the MoS_2 surface while boosting mobility [16].

Most notably, the ferroelectric HfO_2 configuration shows steep-slope behavior with sub-60 mV/dec SS in the subthreshold regime. This steepening is attributed to internal voltage amplification enabled by negative capacitance effects within the ferroelectric dielectric. The resulting memory window — visualized in Fig. 21 — spans approximately 0.6 to 0.8 V, making the structure suitable for non-volatile FeFET applications. In this regime, polarization hysteresis contributes to bistable switching states without the need for additional charge-trap layers, as seen in conventional memory FETs [17], [18].

B. Subthreshold Swing and Threshold Voltage

A primary goal in modern FET design is to minimize the SS, allowing for reduced operating voltage and power consumption. In these simulations, the SS values observed for each gate dielectric configuration were extracted from the linear region of the log(Id)–Vg curves and compared. For the SiO_2-based structure, the SS remained around ~90 mV/dec due to weak capacitive coupling and interfacial traps. With Al_2O_3, the SS was reduced to ~70 mV/dec, indicating stronger gate control and improved interface quality — in line with ALD-grown Al_2O_3's known passivation behavior [19]. The lowest SS, approximately 53 mV/dec, was achieved using ferroelectric HfO_2 under optimal thickness and coercive field conditions, confirming the advantage of negative capacitance engineering [20]. Threshold voltage (V_{th}) was similarly extracted using the constant-current method. The value shifted depending on both the dielectric and the channel doping level.

Notably, ferroelectric-induced charge contributed to V_{th} shifts, consistent with prior experimental studies of MoS_2 NC-FETs [21].

C. Output Characteristics and Saturation Behavior

The Id–Vd characteristics under varying gate voltages were simulated to assess current saturation and drain modulation. These curves (Figs. 4, 5, and 7) display classical FET behavior with distinct linear and saturation regions. With low Vtg, the channel is depleted, and the Id–Vd curve remains flat, showing excellent cutoff behavior. As Vtg increases, current rises sharply in the linear region, eventually reaching saturation. Devices with Al_2O_3 and HfO_2 gate stacks show more pronounced saturation due to enhanced mobility and better electrostatic confinement. The saturation current achieved in the ferroelectric case was highest, reflecting the cumulative benefits of high dielectric strength and polarization-assisted field modulation. The saturation drain current (Idsat) reached values exceeding 100 μA/μm in strong inversion for HfO_2 devices, while SiO_2-based devices saturated below 40 μA/μm under similar bias — demonstrating a clear advantage in performance for high-k and ferroelectric stacks.

D. Carrier Density and Spatial Modulation

The spatial and quantitative evolution of carrier density with respect to gate voltage plays a pivotal role in determining the conduction characteristics of MoS_2 FETs. From Fig. 8, it is evident that electron concentration increases exponentially as the gate bias exceeds the threshold voltage. This transition is sharp and abrupt, reflecting high gate efficiency and confirming that the field effect in ultrathin MoS_2 is exceptionally strong. As shown in Fig. 9 and Fig. 12, both the electric diffusion current density and the electron current density increase dramatically with Vtg. These simulations illustrate how charge carriers begin to redistribute once inversion is initiated. Notably, current density profiles are concentrated near the surface and extend uniformly across the channel width in the strong inversion regime. This implies that the device exhibits minimal short-channel leakage and excellent charge confinement, even as gate bias is increased.

Additionally, the surface current density plot (Fig. 11) shows that current crowding occurs closer to the drain edge at high drain bias. This observation matches experimental expectations and validates the boundary conditions used in the simulation. The peak current density regions identify potential reliability concerns such as electromigration, and provide guidance for layout strategies and current density handling limits in nanoscale devices [22].

E. Recombination Activity in the Channel

Recombination dynamics are a significant factor influencing subthreshold swing, leakage current, and switching energy. The results in **Fig. 16** and **Fig. 17** display the spatially averaged Shockley–Read–Hall (SRH) recombination rates for electrons and holes, respectively, under a range of gate voltages. Near the threshold region, recombination rates peak due to the overlap of minority and majority carrier populations. This is particularly evident in the electron recombination profile, where a bell-shaped curve is observed. As gate bias continues into strong inversion, hole density diminishes, reducing the overall recombination rate. This trend is mirrored in the hole recombination profile (**Fig. 17**), which becomes negligible in the ON-state. These plots provide evidence that trap-assisted recombination is minimal at high gate biases, suggesting good dielectric/channel interface quality. However, the recombination peaks near the switching point may contribute to dynamic power loss and transient charging effects — key concerns in high-speed logic and memory designs [23].

F. Band Alignment and Energy Profiles

Energy band simulation outputs (**Figs. 18** and **19**) offer crucial insight into the internal potential landscape of the MoS_2 FET. In **Fig. 18**, the conduction band energy profile clearly shows the channel barrier lowering as Vtg increases. The band bending is symmetric about the channel center under low Vd, and asymmetric when Vd is raised, resulting in a pinch-off region forming near the drain terminal. This validates the onset of current saturation and supports velocity saturation modeling. **Fig. 19**, which shows the valence band energy, moves only slightly with increasing Vtg — consistent with n-type behavior. The energy gap between conduction and valence bands also shifts slightly due to field-induced band alignment but remains large enough to suppress ambipolar conduction. These results confirm that the gate voltage effectively controls the conduction band minimum while preserving OFF-state integrity through controlled band alignment.

G. Electric Field and Dielectric Performance

The simulated electric field strength (**Fig. 21**) increases steadily with Vtg, peaking near the gate dielectric/channel interface. This peak field must remain below the breakdown threshold of the dielectric stack, especially in thin-layer devices. Al_2O_3 and ferroelectric HfO_2 show higher tolerances to field stress due to their greater dielectric strength. In the ferroelectric case, field localization is also responsible for inducing domain switching in the HfO_2 layer, which plays directly into the hysteresis and memory behavior described below. Thus, understanding electric field evolution is vital for predicting device lifetime, hot-carrier effects, and breakdown susceptibility [24].

H. Ferroelectric Polarization and Hysteresis Window

Perhaps the most distinct signature of the ferroelectric gate stack appears in **Fig. 22**, which depicts the polarization vs. gate voltage behavior of the HfO_2 dielectric. The figure reveals a clockwise hysteresis loop — the hallmark of a ferroelectric FET. This bistable behavior is enabled by spontaneous and switchable polarization states that persist after the external field is removed. The separation between the forward and reverse sweeps constitutes the "memory window," which in this case spans approximately 0.7 V. This confirms that the device can store logic states in a non-volatile manner, a property crucial for applications such as non-volatile memory (NVM), compute-in-memory (CIM), and neuromorphic computing. The sharpness and width of the loop indicate fast switching dynamics and retention stability, which align with experimental benchmarks for doped HfO_2 ferroelectrics. These results suggest that the MoS_2/ferroelectric hybrid structure can perform both logic and memory operations in a single configuration, offering a compact and energy-efficient solution for next-generation circuits [25].

I. Electrostatic and Potential Distribution

Finally, **Fig. 24** displays the electric potential distribution across the channel and dielectric for varying gate voltages. As expected, potential builds up across the oxide stack and drops within the semiconductor body, establishing the field necessary for channel inversion. The potential profile shows no abrupt discontinuities, suggesting smooth band bending and reliable gate-channel interface characteristics. This spatial distribution directly correlates with band diagrams in earlier figures and confirms that voltage control is achieved efficiently and uniformly across the simulated domain. By visualizing how potential evolves across the structure, this plot helps validate the electrostatic boundary conditions and confirms that the modeled field profile supports sharp switching and minimal fringe leakage — critical for ultra-scaled logic designs.

V. OPTIMIZATION AND DESIGN RECOMMENDATIONS

A. Gate Dielectric Engineering

The simulation results presented in previous sections make clear that the selection and engineering of the gate dielectric are pivotal to the performance of MoS_2 field-effect transistors. Among the three dielectric stacks studied — SiO_2, Al_2O_3, and ferroelectric HfO_2 — ferroelectric HfO_2 exhibits superior electrostatic control, subthreshold behavior, and memory functionality. For logic applications requiring ultra-low subthreshold swing (SS < 60 mV/dec), integrating a ferroelectric layer in the gate stack proves highly effective. The negative capacitance effect enabled by HfO_2 offers voltage amplification at the semiconductor interface, thereby reducing switching energy and enhancing the steepness of the subthreshold region. However, to avoid polarization fatigue and instability, the ferroelectric thickness must be optimized — typically within 6–10 nm for doped HfO_2 layers [26].

Al_2O_3, while lacking ferroelectric behavior, provides excellent dielectric integrity and a smooth interface with MoS_2. It is recommended as a non-volatile alternative for devices where hysteresis is undesirable, especially for high-frequency applications. Its moderate κ value balances gate capacitance and leakage current, making it suitable for both logic and analog devices [27]. SiO_2, though widely used in conventional CMOS, demonstrated the weakest gate modulation and highest threshold voltage in simulations. Therefore, it should be limited to initial prototyping or layered beneath high-k materials to form gate stacks with improved EOT (Equivalent Oxide Thickness) characteristics.

B. Channel Thickness Optimization

Channel thickness directly influences both electrostatic control and transport efficiency. Monolayer MoS_2 offers the most aggressive gate coupling and best SS due to its atomic thickness, making it ideal for deeply scaled transistors. However, simulations show that increasing thickness to 2–3 layers moderately improve carrier mobility while slightly degrading gate control. This trade-off must be carefully considered. For high-performance logic circuits where short-channel effects dominate, monolayer or bilayer MoS_2 is recommended to ensure tight control over the channel. In contrast, for analog or memory operations where mobility and current drive are more critical, tri-layer configurations can provide better drive currents at the cost of minor V_{th} shifts [28]. Additionally, multilayer MoS_2 introduces an indirect bandgap that affects tunneling and leakage behavior. Devices using thicker MoS_2 should be paired with higher work function metals or symmetric dual gates to maintain electrostatic integrity.

C. Gate Geometry and Device Architecture

The simulations affirm that top-gated structures provide superior electrostatic control compared to back-gated designs. Top-gate configurations allow for precise EOT control and better field coupling due to proximity to the channel. Therefore, top-gate single- or dual-metal designs are recommended for logic applications. Back-gate designs, while easier to fabricate in lab conditions, suffer from poor threshold tuning and are less scalable. For ultra-scaled applications, gate-all-around (GAA) or dual-gate geometries should be considered. These structures can significantly reduce DIBL and improve saturation behavior, as confirmed in prior experimental and modeling work [29]. Gate length scaling must be coordinated with dielectric and channel optimization. Aggressive reduction below 10 nm requires compensation through high-k dielectrics and contact engineering to suppress parasitic resistances. Lateral scaling should be accompanied by vertical scaling of the oxide and minimized source/drain underlap to mitigate series resistance effects.

D. Memory Integration via Ferroelectric Layers

One of the most promising aspects of the MoS_2 FET design explored here is the integration of ferroelectric memory functionality within the same logic structure. Simulations showed that the use of ferroelectric HfO_2 in the gate stack resulted in a stable polarization hysteresis loop (Fig. 21), with a memory window exceeding 0.6 V. To maximize retention and endurance, it is essential to optimize ferroelectric layer thickness and coercive field. Excessively thick layers can suppress switching, while overly thin layers may suffer from incomplete polarization and domain instability [30]. Additionally, the use of interfacial buffer layers such as TiN or AlN may help improve switching uniformity and suppress wake-up effects commonly seen in ferroelectric oxides. Applications that would benefit most from this design include embedded NVM arrays, low-power edge memory, and neuromorphic circuits where internal state retention without refresh is advantageous. The dual-use of logic and memory in a compact footprint provides a path to logic-in-memory architectures.

E. Reliability, Variability, and Scalability Considerations

Device scaling introduces critical concerns regarding reliability and variability. Based on the field and energy density simulations (**Figs. 11** and **21**), care must be taken to ensure the maximum internal electric field does not exceed dielectric breakdown limits. This is especially crucial in ferroelectric stacks where high coercive fields can induce stress during switching cycles. To minimize variability, especially threshold voltage fluctuations and hysteresis instability, interface engineering remains essential. The interface between MoS_2 and dielectric must be free of traps and fixed charges. Techniques such as atomic layer deposition (ALD) for oxide growth, and surface passivation using sulfur treatment or h-BN layers, have been shown to improve interface quality significantly [31].

From a system-level perspective, variability-aware design must also consider statistical variations in polarization switching, channel doping, and dimensional tolerances. Incorporating compact models extracted from these simulations into circuit-level simulation tools (e.g., SPICE) would allow accurate prediction of performance and reliability across voltage, temperature, and process corners [32].

F. Summary of Recommendations

Parameter	Optimized Choice	Justification
Dielectric	HfO_2 (ferroelectric) or Al_2O_3	Enables steep SS / memory or stable logic performance
Channel Thickness	1–3 layers	Monolayer for SS, trilayer for mobility
Gate Design	Top-gated (dual-gate or GAA)	Stronger electrostatics and better scaling potential
Memory Strategy	FeFET architecture with polarization control	Hysteresis-enabled non-volatile logic-in-memory
Reliability	Field ≤ dielectric breakdown; ALD-grown interfaces	Prevent breakdown; improve interface stability

These recommendations are based on direct physical interpretation of 23 finite-element simulation results and form a foundation for experimental benchmarking and future fabrication efforts.

VI. CONCLUSION

This work has presented a comprehensive finite-element simulation study of MoS_2-based field-effect transistors, focusing on the influence of gate dielectric material, channel thickness, electrostatics, and ferroelectric integration. By evaluating 23 detailed simulation plots across various configurations — including SiO_2, Al_2O_3, and ferroelectric HfO_2 gate stacks — we have developed an in-depth understanding of how each design parameter affects device performance, switching behavior, and potential multifunctionality.

Key findings include the demonstration of sub-60 mV/dec subthreshold swing in ferroelectric gate configurations, confirming the effectiveness of negative capacitance effects for steep-slope logic. Devices with ferroelectric HfO_2 also exhibited clear polarization hysteresis and a memory window of approximately 0.7 V, validating their use in non-volatile memory and logic-in-memory applications. In contrast, Al_2O_3 offered the most stable and reproducible characteristics for pure logic functions, balancing interface quality and dielectric strength. The modeling framework incorporated coupled Poisson, drift-diffusion, and Landau–Khalatnikov equations, enabling accurate prediction of carrier dynamics, recombination rates, energy band alignment, and electric field distributions. This allowed for a highly realistic and quantitative evaluation of device behavior across multiple biasing regimes. Based on the results, we proposed targeted design recommendations for future MoS_2 FETs, suggesting specific combinations of gate materials, channel thicknesses, and structural configurations to balance performance and reliability. These guidelines are expected to support the development of next-generation 2D electronics capable of addressing both logic and memory needs in a compact, energy-efficient form factor.

In future work, this simulation platform may be extended to incorporate time-dependent switching dynamics, temperature effects, and statistical variability for large-scale circuit simulations. The insights and methods developed here establish a foundation for both academic exploration and industrial implementation of MoS_2-based transistor technologies.

VII. REFERENCES

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